

E2 REPORT

MANISH

23B0354

We calculated the Pearson correlation Coefficient (r) for the dataset (used in E1)

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Where,

r = Pearson Correlation Coefficient

x_i = x variable samples

y_i = y variable sample

$\bar{x}$ = mean of values in x variable

$\bar{y}$ = mean of values in y variable

Which gives 0.946181204

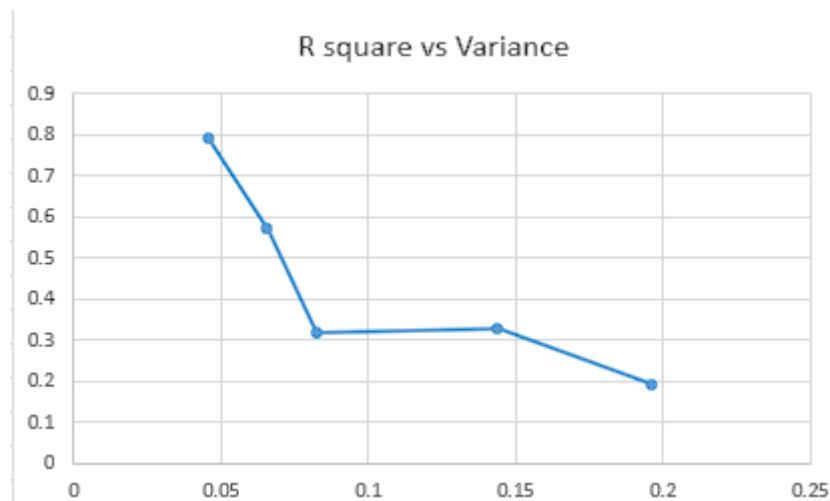
The correlation coefficient tells us about the linear correlation. 0.946 showcases that x and y are highly correlated, which is true since we used x to calculate y previously with some added noise.

Next is the detailed regression output using built-in regression in Excel.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.946596466							
R Square	0.896044869							
Adjusted R Square	0.894973166							
Standard Error	0.06499882							
Observations	99							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	3.532372538	3.532373	836.0948733	1.76925E-49			
Residual	97	0.409810115	0.004225					
Total	98	3.942182653						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.204895312	0.013055303	15.69441	2.26558E-28	0.17898415	0.230806474	0.17898415	0.230806474
	0.474612678	0.615699683	0.021293211	28.91531	1.76925E-49	0.573438555	0.657960812	0.573438555

- The dataset's R2 (The Coefficient of Determination) is 0.896. The R2 shows how good the regression is the closer the R2 is to 1.
0.896 shows that regression here is good.

- R^2 created by regression output = 0.896.
And R^2 calculated using basic definition is 0.895. Both are almost the same.
- Both the R^2 and r^2 are the same because we use the same formula for R^2 and r^2 in linear regression.
- Standard Error is the average distance the observed values fall from the regression line. The standard error in my dataset is 0.06499, which showcases that the precision of my dataset is quite high.
- The p-value here is $2.26E-28$, which is very small; we know from the null hypothesis test that any value below 0.05 is good.
- A 95% confidence interval showcases that we are 95% sure that the actual parameter lies between the intervals. If zero is also included, it suggests that the actual coefficient could be zero, which means the coefficient is not statistically significant.
- F-value is the measure that evaluates the overall significance of a regression.



This shows that R^2 decreases with an increase in variance. This makes intuitive sense since Variance increases the spread of the dataset, and R^2 shows how well the regression line fits.

However, online, I discovered that increasing the variance only by multiplying wouldn't change the R^2 since it's a coefficient.

With this assignment, I learned more about

- Using Excel built-in analytical tool.
- Various statistical parameters
- About Null Hypotheses
- Improved how to communicate my learning